Semester: 5th
Subject Name:-CI & Code:- CS30011
Branch (s): - CSE, IT, CSSE

AUTUMNMID SEMESTER EXAMINATION-2024

School of Computer Engineering
Kalinga Institute of Industrial Technology, Deemed to be University
Computational Intelligence
[CS30011]

Time: 1 1/2 Hours **Full Mark: 20**

Answer Any four questions including question No.1 which is compulsory.
The figures in the margin indicate full marks.
Candidates are required to give theiranswers in their own words as far as practicable and all parts of a question should be answered at one place only.

1. Answer all the questions. [1 X 5]

a) Differentiate between hard computing and soft computing.
b) Consider a two-input single Adaline with the weights [3 2] and inputs [-5 7]^T. Find the bias required to generate an output of 0.5.
c) Test a perceptron with weights $w_1 = 1$, $w_2 = -0.8$ and bias = 0 against the following input/target pairs and update the weights and bias according to the perceptron learning rule.

x_1	x_2	t
1	2	1
-1	2	0
0	-1	0

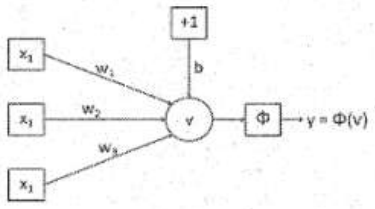
d) Which activation function will you choose in your neuron model if you want a continuous output signal in the range of [-1, +1]? Justify your answer.
e) Explain elitism in GA through an example.

2. [2+3]

(a) Differentiate between ANN and BNN. Sketch the schematic architecture diagram of ANN and specify its components. Write its advantages and disadvantages.

(b) Consider the single McCulloch-Pitts artificial neuron given below. The node receives three-dimensional input patterns $x = (x_1, x_2, x_3)$, each component of which are only binary signals (either 0 or 1). Suppose that the weights corresponding to the three inputs and bias have the following values: $w_1 = 2$, $w_2 = -4$, $w_3 = 1$, $b = 1$, and the activation of the unit is given by the hardlimit / step function:

$$y = \begin{cases} 1, & v \geq 0 \\ 0, & otherwise \end{cases}$$



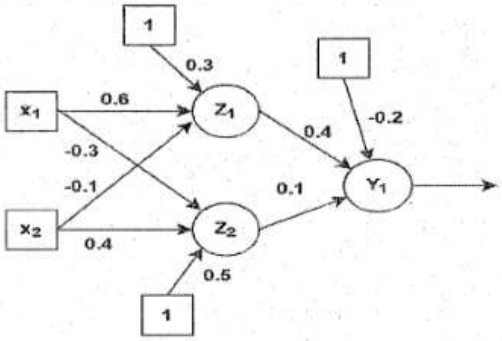
Calculate the output value y of the neuron for each of the following input patterns:

Pattern	x_1	x_2	x_3
P_1	1	0	0
P_2	0	1	1
P_3	1	0	1
P_4	1	1	1

3. [2+3]

(a) What is the role of transfer function in perceptron architecture? State the limitations of single-layer perceptron.

(b) Find the new weights for the following backpropagation network after presenting input [1, -1] once for the target [0]. Use log-sigmoid/Logistic Sigmoid activation function for the hidden and output layers. Assume a learning rate of 0.5.



4. [2+3]

(a) Explain the Adaline model. Implement AND gate using Adaline network with a learning rate of 0.1. Assume initial weights of [0.2 0.3] and an initial bias of 0.1.

(b) Describe with a proper diagram how the RBF network works. Mention its merits and demerits. For which kind of applications would you prefer RBF networks over MLP networks?

5. [2+3]

(a) Briefly explain the characteristics of derivative free optimization.

(b) Apply the steps of GA to maximize the function:

$y = 100 - x^2$ where x is an integer, $-10 \leq x \leq 10$

Assume : (i) rate of crossover to be 100% and rate of mutation to be 0%

(ii) 5-bit binary chromosomes

(iii) population size of 4

Show at least three generations starting from initial population. (Initial population is not a generation.)

*** Best of Luck ***